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Dear Editorial team,

We are excited to submit our manuscript entitled "Artificial intelligences and human scientists exhibit complementary strengths in theory building" for consideration in *Organization Science*, primarily as a regular research article. Given its relevance to the Special Issue on "AI-Enabled Frontiers in Organizational Science," we would also be grateful if the manuscript could be considered for the Special Issue, should the editors find it appropriate. As we have already carried out and written up the results of the project, we are submitting a full manuscript with empirical findings rather than an initial proposal. If there is initial interest on the journal's side, we are of course delighted to adapt the report to any format/length you and the reviewers prefer.

The manuscript reports on a crowd science initiative competing 73 human scholars (including 13 senior scientists and 60 doctoral students) and 25 generative AI models at advanced theory-building tasks. Provided with detailed descriptions of sets of independent and dependent variables regarding race and gender inequality discourses, human scientists and GenAI systems attempted to develop *a priori* theoretical models and predict relevant empirical relationships. Then, after learning about the realized findings, they provided *post-hoc* theoretical explanations for the observed patterns, revising or defending their initial theoretical models. A crowdsourced approach allowed us to compare human scientists and artificial intelligences at both the individual and collective levels.

Key findings:

- The average AIs outperformed the average human on most of the theory building tasks and did especially well at anticipating some complex interaction effects.
- Human theories were more diverse and exhibited greater gains in predictive accuracy from aggregation.
- AI theories were extensively elaborated, involving additional theoretical paths and latent variables, and were rated as higher quality than human theories by independent raters blinded to source. However, this theoretical complexity was in part ornamental, in that it was not associated with more accurate scientific predictions about correlational empirical patterns in data; in contrast, human scientists achieved greater predictive efficiency with simpler theories.

- AIs were significantly more likely than human scientists to revise their theories to incorporate new evidence; human scientists updated their beliefs in a selective manner that is sensitive to prior prediction errors.

These complementary comparative advantages point to promising avenues for human–AI collaboration in scientific settings. We speculate that the superior processing capacity of artificial intelligences may makes them especially well-suited to tasks requiring grappling with complexity, yet the greater diversity of human ideas could be essential to wise crowds and collective creativity. These are of course initial findings that await more systematic tests by the scientific community. Our materials are fully open source, and we encourage further investigations leveraging the paradigm we introduce across diverse research domains and with increasingly advanced artificial intelligences.

The manuscript is original and not under consideration elsewhere. A repository with reproduction materials is available at https://github.com/KeLiChloe/human-ai-theory-building. The pre-registration of our analysis plan is available here: https://osf.io/t7hmu/overview?view_only=e8077497caff4f2384275c778c993efc.

Notably, members of the crowd of scientists who contributed theories regarding race and gender inequality discourses are listed as co-authors on the enclosed manuscript, accounting for the long author string. These contributors, who as noted included a number of senior topic experts, both formulated theories as part of the crowd initiative and helped write and edit the manuscript itself, thus qualifying for authorship.

Our recommended reviewers include the following experts on AI-powered and big data social science:

- **Michal Kosinski**, Stanford University, michal@michalkosinski.com
- **Tessa Charlesworth**, Northwestern University, tessa.charlesworth@kellogg.northwestern.edu
- **Gilad Feldman**, Universität Innsbruck, giladfel@gmail.com,
- **David Rand**, Cornell University, rand.dave@gmail.com
- **Michael Frank**, Stanford University, mcfrank@stanford.edu

Thank you for your consideration.

The authors

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